

Saransh Agrawal

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Education

Texas A&M University

Aug 2023 – May 2025

Master of Science in Data Science: GPA: 3.7/4.0

College Station, TX

- Research Assistant at FLAIR Lab - Responsible NLP, Safety, Robustness. 2024 - Present

- Teaching Assistant, Dept. of Statistics. Fall 2024

Manipal Institute of Technology

July 2017 – Aug 2021

Bachelor of Technology in Control Systems Engineering.

Manipal, IN

Technical Skills

Languages: Python, C/C++, SQL, Node.js, Unix/Shell.

Tools/Frameworks: Transformers, PyTorch, TensorFlow, vLLM, Gymnasium, Scikit-Learn, Numpy, OpenCV, NLTK.

Devops and Misc.: Kafka, Docker, Keras, Jenkins, Kubernetes, Flask, FastAPI, Dask, PyQT, Git, AWS, Matplotlib, Plotly, Pandas, MongoDB, Elasticsearch, PostgreSQL, OpenMP.

Coursework: NLP, Machine Learning, Reinforcement Learning, Parallel Computing, Data Mining, Optimization, Computational Big Data, Data Streaming and Algorithms, DSA.

Publications

- **Saransh Agrawal** and Kuan-Hao Huang. SHA256 at SemEval-2025 Task 4: Selective Amnesia – Constrained Unlearning for Large Language Models via Knowledge Isolation. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval 2025 at ACL)*.

Projects

- **Multi-Preference Alignment via Task Vectors:** Designed a novel pipeline merging single-preference models via task vectors to balance helpfulness/harmlessness in LLMs. (*preprint*)
 - Outperformed **Safe-RLHF** with lower compute cost and enabled precise post-hoc control. *Submitted to COLM 2025.*
- **Boosting MMLLM Long-Context Performance via Cache Merging:** Addressed **KV cache** bottlenecks limiting **Multimodal LLMs (LLaVA, MiniGPT4)** on **MileBench** by implementing **Cache Merging (CaM)**.
 - Engineered technique to significantly improve memory efficiency, enabling robust reasoning over extended sequences.
- **Always-On-Policy Prompts for Efficient RLHF:** Developed **performance-adaptive** prompt generation strategy for **PPO**, dynamically adapting training data based on model scores to improve efficiency.
 - Outperformed static methods in helpfulness alignment (**GPT-2**) under identical compute budgets.
- **Scalable LLM Benchmarking Framework for HPC (SLURM):** Developed framework simplifying distributed **LLM** benchmarking (**MMLU**) on **HPC** systems.
 - Engineered reusable **PyTorch DDP/Accelerate** solution with **SLURM** templates enabling evaluation across multi-node, multi-GPU configs on **HPRC**.
- **20 Questions Game Agent:** Fine-tuned **Gemma 2/Llama 3.1** using **RL (PPO)**, **LoRA** for expert-level gameplay.
 - Generated on-the-fly dataset and used a custom reward function to enhance strategic questioning/inference.
- **Crop Monitoring System & Yield Prediction:** Architected a **microservice-based** web platform using satellite imagery (Sentinel/PlanetScope via **QGIS**).
 - Fine-tuned **ViT/Satlas** on temporal data for yield prediction. Presented at *ASA Conf. 2024*.

Work Experience

Software Developer, ML Team

July 2021 – July 2023

Viewzen Labs

Bangalore, India

- Conceptualized and spearheaded the development of novel **AutoML platform (MLaaS)**, for supervised learning.
- Developed **System Architecture** by orchestrating **microservices** through **REST APIs** and **Kafka Streams**.
- Developed a **Python Library** that implements the AutoML logic, using **Hyperopt** and **Scikit-Learn**, reducing the pre-processing, model development and evaluation time by **80%**.
- Developed multiple RESTful Python microservices, containerized using **Docker** and ingesting data through **asynchronous Kafka consumers**. Utilized **multiprocessing** to handle large volume of requests.
- Crafted Python-based microservice for **big data transformation** in **ETL (SaaS) product**. Leveraged **Cython**, to improve CPU utilization and enhanced processing speeds by **20%** over the previous Node.js build.
- Led a large-scale initiative predicting pregnancy complications, encompassing **data processing, analysis, and predictive modeling** for 45,000+ women (300+ features), culminating in a medical desktop application.

Machine Learning Engineer Intern

Feb 2021 – June 2021

Centre for Digital Financial Inclusion

Delhi, India

- Transformed raw audio signal to its MFCC components using **custom TensorFlow** layers.
- Implemented **RNN, GRU and LSTM** architectures in predicting text at character, word and phoneme level.
- Utilized **DeepSpeech** model for **transfer learning** on **Indian Accent dataset** to achieve a result of **0.10 CER** and **0.17 WER** with a **context-aware KenLM** language model. Added **T5 Transformer** to generate **SQL queries**.